Change of Numeraire

Choosing the unit that makes pricing simple

Lecture 11

One price, many useful representations

The question

Risk-neutral pricing discounts by the money-market account. What changes if a different positive traded asset is used as the unit of account?

1. Define a numeraire and the measure that comes with it.
2. Derive the pricing formula in any unit of account.
3. See why a bond numeraire produces a forward measure.
4. Apply the same logic to domestic and foreign currency viewpoints.

NO-ARBITRAGE The no-arbitrage price does not change. We change only its representation: the unit, the measure, and the martingale process.

Units, measures, and martingales

Risk-neutral valuation already uses a numeraire

With money-market account B_t, the usual pricing formula is

$$V_t = B_t \, \mathbb{E}^{\mathbb{Q}} \left[\frac{X_T}{B_T} \mid \mathcal{F}_t \right].$$

Equivalently, for a constant rate r,

$$V_t = \mathbb{E}^{\mathbb{Q}} \left[e^{-r(T-t)} X_T \mid \mathcal{F}_t \right].$$

What is a numeraire?

It is the strictly positive traded asset used as the unit in which prices are quoted. Here, one unit means one unit of cash invested in B.

NO-ARBITRAGE Under the measure associated with B, every traded price divided by B_t is a martingale.

A numeraire is a change of unit, not a new economy

Let $N_t > 0$ be a traded asset. Its denominated price for an asset S is

$$S_t^{(N)} = \frac{S_t}{N_t}.$$

Choice of unit	Martingale object	Associated measure
Money-market account B_t	S_t/B_t	$\mathbb{Q}$
Zero-coupon bond $P(t,T)$	$S_t/P(t,T)$	$\mathbb{Q}^T$
Stock S_t	X_t/S_t	$\mathbb{Q}^S$

WATCH OUT Dividing by N_t but keeping the old measure is generally wrong. The numeraire and probability measure must change together.

Ratios carry an Itô correction

Suppose, under one measure,

$$\frac{\mathrm{d}S_t}{S_t} = \mu_S\,\mathrm{d}t + \sigma_S\,\mathrm{d}W_t, \qquad \frac{\mathrm{d}N_t}{N_t} = \mu_N\,\mathrm{d}t + \sigma_N\,\mathrm{d}W_t.$$

Then the ratio satisfies

$$\frac{\mathrm{d}(S_t/N_t)}{S_t/N_t} = \left(\mu_S - \mu_N + \sigma_N^2 - \sigma_S\sigma_N\right)\mathrm{d}t + (\sigma_S - \sigma_N)\,\mathrm{d}W_t.$$

YOUR TURN Is the drift of S_t/N_t simply $\mu_S - \mu_N$?

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No

The quadratic-variation terms are real. The change of measure is precisely what removes the remaining drift of the denominated price.

Every positive traded numeraire has its own martingale measure

Start under the money-market measure $\mathbb{Q}$. The measure associated with a positive traded numeraire N is defined by

$$\left.\frac{\mathrm{d}\mathbb{Q}^N}{\mathrm{d}\mathbb{Q}}\right|_{\mathcal{F}_t} = \frac{N_t/B_t}{N_0/B_0}.$$

Why this is a valid density

N_t/B_t is a positive $\mathbb{Q}$-martingale because N is traded. After normalizing at time 0, its expectation is one.

NO-ARBITRAGE Under $\mathbb{Q}^N$, every traded asset price divided by N_t is a martingale.

Changing measure reweights states; it does not create states

For two positive numeraires N and M,

$$\left.\frac{\mathrm{d}\mathbb{Q}^M}{\mathrm{d}\mathbb{Q}^N}\right|_{\mathcal{F}_t} = \frac{M_t/N_t}{M_0/N_0}.$$

Economic reading

States in which M performs well relative to N receive more weight under $\mathbb{Q}^M$ than under $\mathbb{Q}^N$.

What stays fixed

The measures are equivalent: they assign zero probability to exactly the same events.

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Answer

No. It changes probability weights, not the scenarios themselves.

The general pricing formula has the same three steps

For a claim paying X_T at time T,

$$\boxed{V_t = N_t \, \mathbb{E}^{\mathbb{Q}^N}\left[\frac{X_T}{N_T} \mid \mathcal{F}_t\right].}$$

1. Divide the payoff by the numeraire at the payment date.
2. Take conditional expectation under the associated measure.
3. Multiply by the current numeraire value.

NO-ARBITRAGE A good numeraire makes the normalized payoff X_T/N_T simple and makes the relevant normalized price a martingale.

The money-market formula is one special case

Set $N_t = B_t$. Then $\mathbb{Q}^N = \mathbb{Q}$ and

$$V_t = B_t \, \mathbb{E}^{\mathbb{Q}} \left[\frac{X_T}{B_T} \mid \mathcal{F}_t \right].$$

For deterministic r,

$$\frac{B_t}{B_T} = e^{-r(T-t)},$$

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YOUR TURN What is the density process when the new numeraire is again B_t?

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Answer

It is identically one. Choosing the money-market account does not change the usual risk-neutral measure.

Forward measures simplify payment-date pricing

A zero-coupon bond absorbs discounting

Fix payment date T and choose the bond $P(t,T)$ as numeraire. Since $P(T,T)=1$,

$$\boxed{V_t = P(t,T)\,\mathbb{E}^{\mathbb{Q}^T}[X_T \mid \mathcal{F}_t].}$$

Under $\mathbb{Q}$

Discount factor and payoff may both be random:

$$V_t = \mathbb{E}^{\mathbb{Q}}\left[e^{-\int_t^T r_s\,\mathrm{d}s} X_T \mid \mathcal{F}_t\right].$$

Under $\mathbb{Q}^T$

The discounting is embedded in $P(t,T)$; only the payoff remains in the expectation.

Forward prices are martingales under their payment-date measure

For an asset with no intermediate cash flows, define

$$F(t,T) = \frac{S_t}{P(t,T)}.$$

Under the T-forward measure,

$$\boxed{F(t,T) = \mathbb{E}^{\mathbb{Q}^T}[S_T \mid \mathcal{F}_t].}$$

What this buys us

The forward price has zero drift under the measure naturally attached to the delivery date. This is often much cleaner than tracking a stochastic short rate under the money-market measure.

A forward contract is priced by one martingale condition

A forward on a non-dividend-paying asset delivers one share at T for delivery price K. Its value is

$$V_t = S_t - KP(t,T) = P(t,T)\big(F(t,T) - K\big).$$

YOUR TURN If $S_0 = 100$ and $P(0,T) = 0.95$, what delivery price makes the forward worth zero at inception?

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Answer

Set $K = F(0,T) = S_0/P(0,T) = 105.26$. The forward measure did not change the economic price; it exposed the natural martingale behind it.

Choose the numeraire that cancels a payoff factor

Suppose a claim has payoff $X_T = S_T Y_T$. Under the stock numeraire,

$$\boxed{V_t = S_t \, \mathbb{E}^{\mathbb{Q}^S}[Y_T \mid \mathcal{F}_t].}$$

The factor S_T has disappeared from the expectation.

Good use case

Payoffs naturally proportional to the terminal stock, such as exchange options or share-settled claims.

Condition

The candidate must be a strictly positive traded asset. A payoff that can hit zero is not automatically a valid numeraire.

A practical numeraire-selection checklist

1. What is the payoff date and currency?
2. Does the payoff contain a positive traded factor?
3. Does a bond maturing on the payment date remove discounting?
4. Under which measure is the relevant ratio a martingale?
5. Can the final answer be checked against money-market pricing?

YOUR TURN A payoff is paid at T and depends on future bond prices. Which first candidate for the numeraire should you test?

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Answer

The zero-coupon bond $P(t, T)$. It matches the payment date and generates the T-forward measure.

FX makes the two viewpoints concrete

An exchange rate links two money-market accounts

Let X_t be domestic currency per unit of foreign currency. With constant rates,

$$B_d(t) = e^{r_d t}, \qquad B_f(t) = e^{r_f t}.$$

Object	Units	Domestic value
One foreign currency unit	foreign	X_t domestic
Foreign bank account	foreign	$\boxed{X_t B_f(t)}$ domestic
Domestic bank account	domestic	$B_d(t)$ domestic

NO-ARBITRAGE $X_t B_f(t)$ is a positive traded asset in domestic currency. It is the natural bridge between domestic and foreign pricing viewpoints.

Domestic no-arbitrage fixes the FX drift

Under the domestic money-market numeraire, the discounted domestic value of foreign cash must be a martingale:

$$\frac{X_t B_f(t)}{B_d(t)} \quad \text{is a } \mathbb{Q}^d\text{-martingale.}$$

Therefore

$$\boxed{\frac{\mathrm{d}X_t}{X_t} = (r_d - r_f)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{Q}^d}.}$$

Interpretation

The foreign rate acts exactly like a continuous dividend yield. Holding foreign currency gives access to foreign interest, just as holding a stock can generate dividends.

The FX forward is the first diagnostic

For delivery of one unit of foreign currency at T,

$$\boxed{F_{0,T} = X_0 e^{(r_d - r_f)T}.}$$

YOUR TURN Take $X_0 = 1.10$ USD/EUR, $r_d = 5\%$, $r_f = 3\%$, and $T = 1$. What is the no-arbitrage one-year forward rate in USD/EUR?

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Answer

$$F_{0,1} = 1.10 e^{0.02} = 1.1222 \text{ USD/EUR}.$$

Domestic interest pushes the forward up; foreign interest pushes it down.

An FX call is Black–Scholes with a foreign yield

A domestic-currency call on one unit of foreign currency pays $(X_T - K)^+$. Its domestic value is

$$\boxed{C_t = X_t e^{-r_f \tau} \Phi(d_1) - K e^{-r_d \tau} \Phi(d_2),}$$

where $\tau = T - t$ and

$$d_1 = \frac{\ln(X_t/K) + (r_d - r_f + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \qquad d_2 = d_1 - \sigma\sqrt{\tau}.$$

NO-ARBITRAGE Read this as the usual dividend-paying-stock formula: domestic rate discounts the strike, foreign rate discounts the spot.

Worked FX option: identify the two discount factors

Let $X_0 = 1.10$, $K = 1.12$, $r_d = 5\%$, $r_f = 3\%$, $\sigma = 12\%$, and $T = 1$. Then

$$d_1 = 0.077, \qquad d_2 = -0.043,$$

and

$$C_0 = 1.10e^{-0.03}\Phi(0.077) - 1.12e^{-0.05}\Phi(-0.043) \approx 0.052.$$

Reading the answer

The option costs approximately 0.052 USD per EUR of notional. The two rates appear in different places because the contract exchanges two currencies.

FX put–call parity reveals the same structure

For domestic prices C_t and P_t with the same K, T,

$$\boxed{C_t - P_t = X_t e^{-r_f(T-t)} - Ke^{-r_d(T-t)}.}$$

YOUR TURN Which term is the present value of one unit of foreign currency received at T: $X_t e^{-r_f(T-t)}$ or $X_t e^{-r_d(T-t)}$?

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Answer

$X_t e^{-r_f(T-t)}$. Invest one foreign unit at the foreign rate, then convert its future value into domestic currency.

Domestic and foreign measures are linked by the foreign bank account

In domestic currency, choose

$$N_t^f = X_t B_f(t)$$

as the foreign-cash numeraire. The density from the domestic measure is

$$\boxed{\left.\frac{\mathrm{d}\mathbb{Q}^f}{\mathrm{d}\mathbb{Q}^d}\right|_{\mathcal{F}_t} = \frac{X_t B_f(t)/B_d(t)}{X_0 B_f(0)/B_d(0)}.}$$

The two viewpoints

Domestic pricing measures values in domestic cash; foreign pricing measures values relative to foreign cash. They give the same no-arbitrage price after the appropriate currency conversion.

A change of numeraire is useful only with the matching measure

Pricing situation	Helpful numeraire	Martingale quantity
Ordinary equity claim	Money-market account	S_t/B_t
Payoff at one fixed date	Bond $P(t,T)$	$S_t/P(t,T)$
Payoff proportional to stock	Stock S_t	X_t/S_t
FX in domestic currency	Domestic cash B_d	$X_t B_f(t)/B_d(t)$

WATCH OUT Never mix the numerator from one convention with the measure from another. That is the most common source of sign and drift mistakes.

Takeaways

The lecture reduces to one reusable formula

$$\boxed{V_t = N_t \, \mathbb{E}^{\mathbb{Q}^N}\left[\frac{X_T}{N_T} \mid \mathcal{F}_t\right].}$$

1. A numeraire is a positive traded unit of account.
2. It determines an equivalent measure under which denominated prices are martingales.
3. The price is invariant; the representation can become much simpler.
4. Bond numeraires create forward measures; FX supplies a natural two-currency application.

Final check: choose the unit before doing the algebra

YOUR TURN A claim pays at date T in domestic currency and its payoff is a function of a T-maturity forward rate. What numeraire is the natural first choice, and what is the resulting pricing measure?

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YOUR TURN A claim pays at date T in domestic currency and its payoff is a function of a T-maturity forward rate. What numeraire is the natural first choice, and what is the resulting pricing measure?

Answer

Start with the zero-coupon bond $P(t, T)$. It leads to the T-forward measure and removes discounting from the expectation at the payment date.

The numeraire changes the view, not the value.